Life Cycle Assessment for Galaxy Book5 Pro 14(UK)

Background

Samsung has developed the technical expertise to analyze and evaluate the environmental impact of its products. Based on international standards such as ISO14040 and ISO14044, Samsung has implemented the SDP*. This evaluation considers the entire product lifecycle, including manufacturing, distribution, usage, and disposal. The SDP analyzes various data categories, including the materials and weight of product BOM**, inputs/outputs in manufacturing processes, distribution routes, energy consumption during product usage, and disposal scenarios to precisely measure environmental impacts.

The LCA results identified and quantified 11 environmental impact categories, including climate change, acidification, ecotoxicity, and ozone depletion. Each category was evaluated for every lifecycle stage. These results will be utilized to improve product environmental specifications and inform product development.

This verification includes implementation methods, related procedures, and requirements for LCA, but does not ensure the reliability of the data used for the product model or the resulting outcomes. Considering these uncertainties, this report will be continuously updated and improved.

Calculation basis

Standard	ISO 14040:2006 and 14044:2006
Database	Ecoinvent 3.10
Method for impact assessment	Life cycle impact assessment classification and characterization factors according to CML v4.8 (Climate Change:IPCC)
LCA software	SDP(Sustainability Data Platform)

SDP* : Sustainability Data Platform

BOM** : Bill of Material

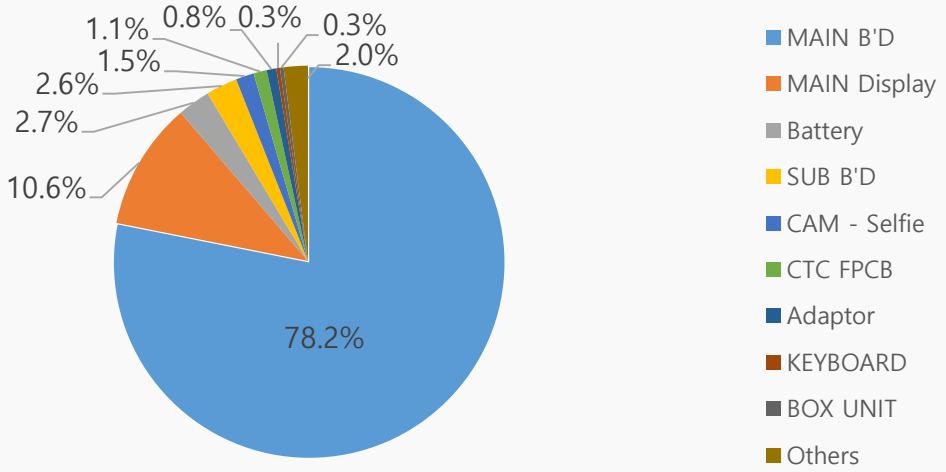
System boundary of LCA

Pre-manufacturing	Parts and materials constituting the products and its transportation
Manufacturing	Product assembly by Samsung Electronics
Distribution	From Vietnam to UK
Use	4 years use
Disposal	Waste treatment of parts and material

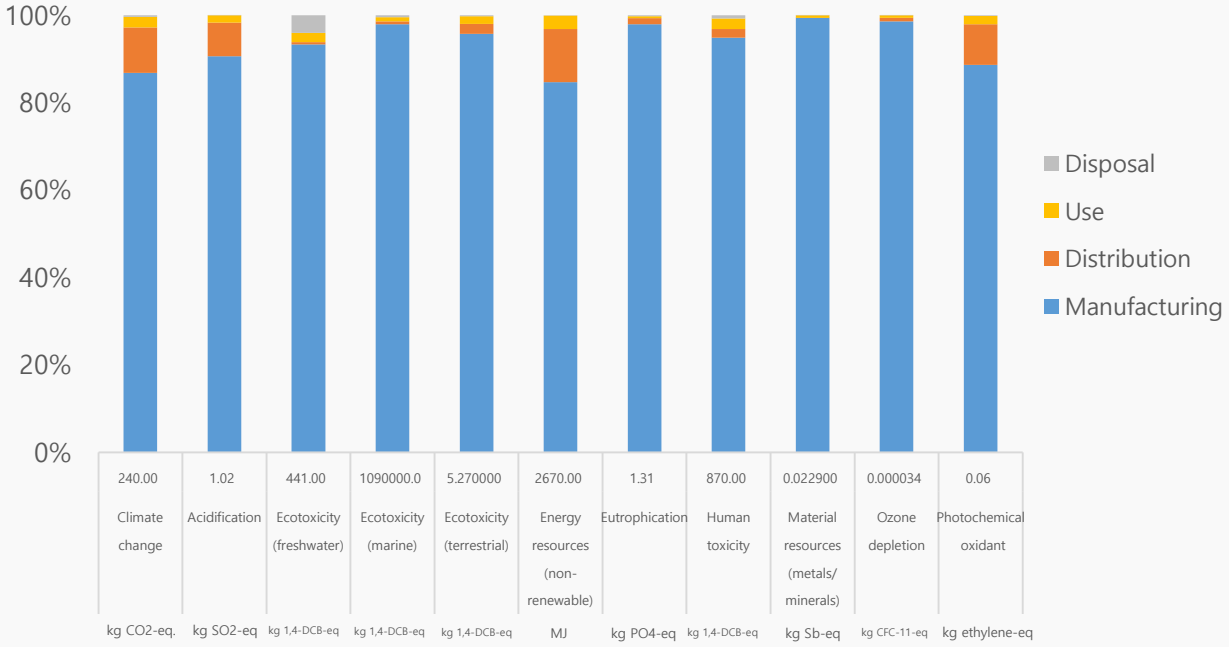
Product Features

Model name	NT940XHA (Galaxy Book5 Pro14)	
Dimension (mm)	312.3 x 223.8 x 11.6	
Display (mm)	355.6	
Weight (g)	Product & Acc.	1415.16
	Packages	857.58

Global Warming Impact Profile

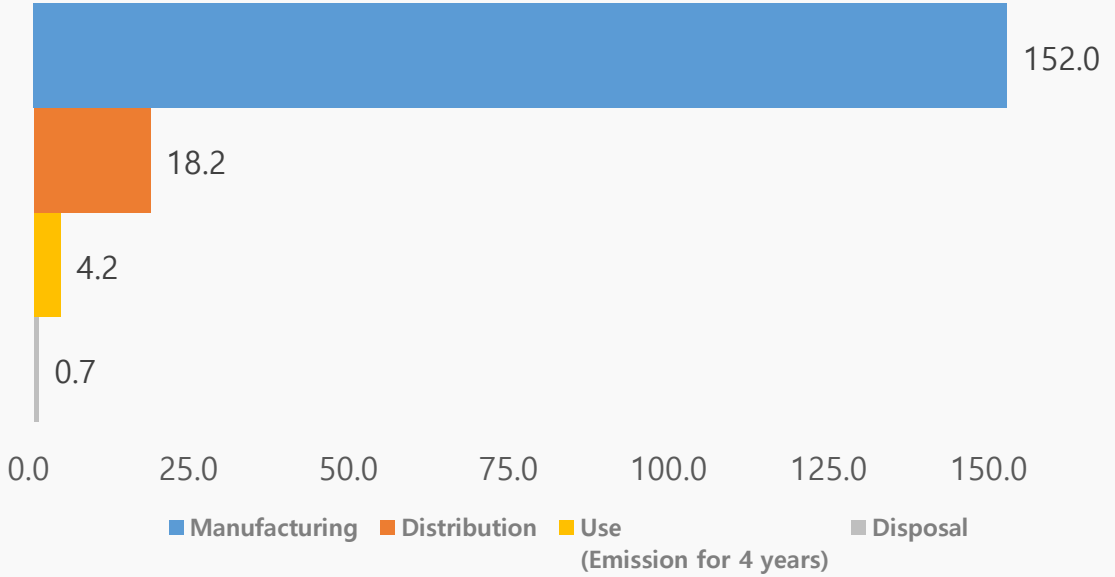


Characterized Environment Impact



Life Cycle Carbon Emissions

Unit : $kgCO_2$ eq.



* The results differ from to region, But not by much.